NUMERICAL RESULTS

Fictional Lab
MEDICAL TEST LABORATORY
EUROPE
FLE

Fictional Lab Europe
Site A – 10 Example Street, 99999 Sampletown
Site B – 5 Placeholder Avenue, 99998 Mocktown

Mrs Jeanne EXEMPLE – born 01/01/1990 – F
Prescriber: Dr Martin TEST
Ref. FLE-250618-0731-TEST – Sample date: 18/06/2025 – Report date: 27/06/2025

GENERAL INFORMATION

Haemolysis index	Normal		
Icterus index	Normal		
Lipemic index	Normal		

HAEMATOLOGICAL BIOCHEMISTRY

Iron	92	60 - 170	µg/dL
Ferritin	38	30 - 200 Health value: 50 - 200	µg/L
Transferrin	2.61	2.00 - 3.60	g/L
Transferrin saturation	25.0	20 - 45	%

OXIDATIVE STRESS STATUS

Vitamins

25-OH Vitamin D	↘ 62	75 - 250	nmol/L

Trace elements

Zinc	13.4	11.0 - 19.0	µmol/L
Copper	↘ 12.1 Alternative units: 76.9 µg/dL Reference values, men: 70.0 - 140.0 µg/dL Reference values, women: 80.0 - 152.0 µg/dL	12.6 - 24.0	µmol/L
Copper	↘ 76.9	80 - 152	µg/dL
Selenium	↗ 131	70 - 120	µg/L

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SALIVARY HORMONOLOGY

Cortisol at wake-up	8.9	5.0 - 18.0	nmol/L
Cortisol wake-up + 30'	16.2	8.0 - 26.0	nmol/L
Cortisol 12h	3.6	1.8 - 5.5	nmol/L
Cortisol 16-20h	↗ 3.9	0.4 - 3.2	nmol/L
Cortisol 22h	0.9	0.3 - 1.5	nmol/L

Results validated by: Dr Alice FICTIVE (Site A) – Dr Paul PLACEHOLDER (Site B)

COMMENTS

Oxidative stress status

FOREWORD

Your patient has had an oxidative stress assessment. This panel looks at the antioxidant vitamins and trace elements that support the body's defences against free radicals, together with a salivary cortisol curve describing the daily rhythm of the stress axis. Each result should be read in the light of the clinical picture, diet and any supplements already taken.

When all markers lie within their reference ranges, defences are usually considered balanced. One or more values outside these ranges suggest an imbalance that may justify dietary adjustments or targeted supplementation, followed by a check after about three months.

Disturbed profile:

Your patient's assessment shows four results outside the reference ranges (vitamin D, copper, selenium and late-afternoon cortisol). Detailed comments are given below for each of them.

About the nutritional advice:

The advice and food tables below are general guidance. Amounts are given per 100 g of food and are indicative only; a more complete table is available on request.

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25-OH Vitamin D ↘

Physiology: *Vitamin D is produced mainly by the skin under sunlight and, to a lesser extent, provided by food. It regulates calcium and phosphate absorption, contributes to bone strength and also plays a part in muscle function and immunity.*

Decreased vitamin D: Your patient's 25-OH vitamin D level is below the reference range (62 nmol/L). This insufficiency is common at the end of winter and in people with little sun exposure. It may be associated with fatigue, muscle weakness and reduced bone mineralisation over time.

Nutritional and micronutritional advice: It is recommended to *increase* moderate sun exposure and the consumption of oily fish and eggs. A cholecalciferol (vitamin D3) supplement of 1,000 to 2,000 IU per day may be considered, with a check after 3 months. Ten foods rich in vitamin D (µg per 100 g) are listed below.

Cod liver oil	**250.0**	Smoked herring	**20.0**
Wild salmon	**12.0**	Sardines in oil	**8.5**
Rainbow trout	**7.0**	Mackerel	**6.0**
Egg yolk	**5.0**	Canned tuna	**3.5**
UV-exposed mushrooms	**3.0**	Fortified margarine	**2.5**

Indicative values, fictitious table for test purposes.

References

1. Doe J, Sample A. Seasonal variation of vitamin D status in an imaginary population. Journal of Test Medicine 2019;12:34-40. (fictitious reference)

Cu ↘ (Copper)

Physiology: *Copper is a trace element required by several enzymes involved in energy production, iron transport, connective tissue formation and antioxidant protection, notably as a cofactor of superoxide dismutase.*

Decreased copper: Your patient's copper level is slightly below the reference range (12.1 µmol/L, i.e. 76.9 µg/dL). A low copper status may weaken antioxidant defences and impair iron use. An excessive zinc intake should be ruled out, as zinc reduces copper absorption.

Nutritional and micronutritional advice: It is recommended to *increase* the consumption of copper-rich foods such as offal, shellfish, nuts and pulses. If needed, a supplement of about 1 mg of copper per day, taken away from zinc, may be prescribed with a check after 3 months. Ten foods rich in copper (mg per 100 g) are listed below.

Beef liver	**12.0**	Oysters	**4.5**
Cocoa powder	**3.8**	Cashew nuts	**2.1**
Sesame seeds	**1.6**	Sunflower seeds	**1.5**
Hazelnuts	**1.4**	Dark chocolate	**1.2**
Buckwheat	**1.1**	Cooked lentils	**0.3**

Indicative values, fictitious table for test purposes.

References

1. Roe R, Example B. Zinc–copper interactions in an invented cohort. Placeholder Clinical Nutrition 2021;8:101-109. (fictitious reference)

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Se ↗ (Selenium)

Physiology: *Selenium is a trace element built into selenoproteins such as glutathione peroxidase. It contributes to antioxidant defences, to thyroid hormone metabolism and to immune function. The margin between adequate and excessive intake is narrow.*

Excess: Your patient's selenium level is above the reference range (131 µg/L). This most often results from supplements or a very high intake of Brazil nuts. A persistent excess may cause digestive discomfort, brittle nails and hair loss.

Nutritional and micronutritional advice: It is recommended to *reduce* the consumption of the richest foods listed below and to *stop any supplement containing selenium*, with a check after 3 months. Ten foods rich in selenium (µg per 100 g) are listed below.

Brazil nuts	**1900.0**	Pork kidney	**190.0**
Tuna	**90.0**	Sardines	**50.0**
Shrimp	**38.0**	Cod	**33.0**
Eggs	**30.0**	Sunflower seeds	**53.0**
Chicken breast	**22.0**	Button mushrooms	**9.0**

Indicative values, fictitious table for test purposes.

References

1. Smith Q, Mock L. Selenium supplementation and blood levels in a fictional trial. Annals of Example Research 2018;5:77-83. (fictitious reference)

Cortisol 16-20h ↗

Physiology: *Cortisol follows a daily rhythm: it peaks about 30 minutes after waking, then falls steadily during the day to reach its lowest level at night. This curve helps the body adapt to stress and supports restful sleep.*

Excess: Your patient's late-afternoon salivary cortisol is above the reference range (3.9 nmol/L), while the morning and night values are normal. This pattern may reflect stress or stimulation in the second half of the day and may delay falling asleep.

Nutritional and micronutritional advice: It is recommended to *limit* caffeine after 2 pm, to keep regular meal and bedtime hours and to favour a calm evening routine. A magnesium supplement may be considered. Ten common sources of caffeine (mg per 100 mL or 100 g) are listed below.

Espresso	**210.0**	Cocoa powder	**230.0**
Filter coffee	**40.0**	Energy drink	**32.0**
Black tea	**20.0**	Dark chocolate	**80.0**
Green tea	**12.0**	Cola	**10.0**
Yerba mate	**15.0**	Milk chocolate	**20.0**

Indicative values, fictitious table for test purposes.

References

1. Brown K, Dummy S. Evening salivary cortisol and caffeine intake in invented volunteers. Imaginary Journal of Endocrinology 2020;3:15-22. (fictitious reference)

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